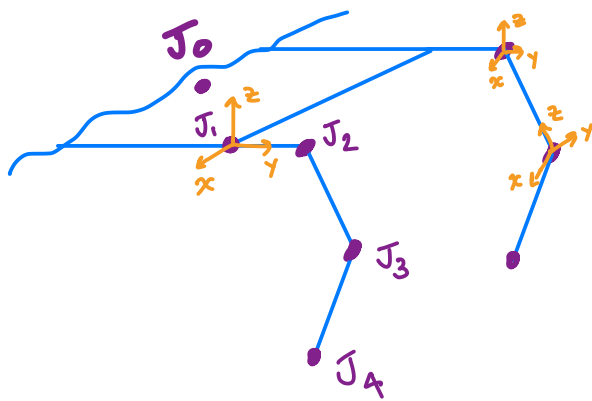


Co-ordinate System:



J_0 : center of Dog [CoM]

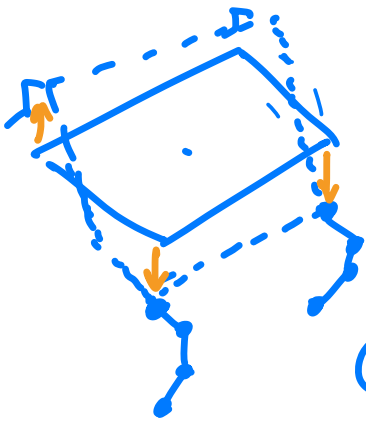
J_1 : Leg origin

J_2 : Femur origin

J_3 : Tibia origin

J_4 : foot

Orientation Control / Rotating Main Body:



① New co-ord of feet relative to J_0 , center of Robot:

$$\overrightarrow{XYZ}_0 = R^{-1} \left(\underbrace{[CoM \text{ origin}]}_{\text{location of } J_1 \text{ relative to } J_0} + \underbrace{[x, y, z]}_{\text{location of foot, relative to } J_1} - \underbrace{[Center \text{ of Rotation}]}_{\text{JR, point to rotate about.}} \right)^T$$

$R = R_{\text{yaw}} R_{\text{pitch}} R_{\text{roll}}$, desired rotation matrix

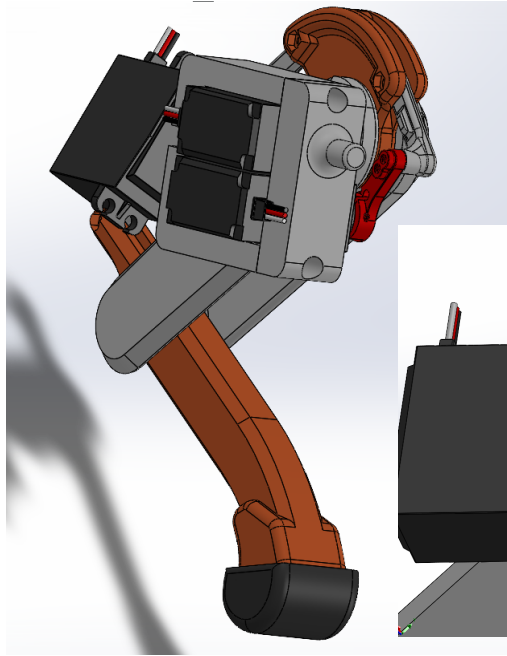
$= [x \ y \ z]^T \leftarrow \text{column vector}$

② New co-ord of feet relative to leg's origin:

$$\overrightarrow{XYZ}_1 = \overrightarrow{XYZ}_0 - [CoM \text{ origin}] + [Center \text{ of Rotation}]$$
$$= [x', y', z']^T \leftarrow \text{Use Inverse Kinematics on this.}$$

Inverse Kinematics: Known: Foot (end effector) position

Desired: Joint angles: θ_{hip} , θ_{femur} , θ_{tibia}



Reference Dimensions:

Diagram illustrating a prosthetic arm mechanism with various dimensions and angles. Key dimensions and angles shown include:

- Dimensions: 29.46, 24.00, 48.00, 24.00, 40.50, 109.50, 109.50, 40.50, 119.90, 5.88°.
- Angles: 87.08°, 92.96°, 174.12°.
- Handwritten annotations: "varies", "fixed angle", "desired foot contact point", "L2", "L3", "L4?", "Do I need to".

→ $\therefore$ tibia cunes, this point, the joint, and desired foot contact point are not collinear

fixed angle

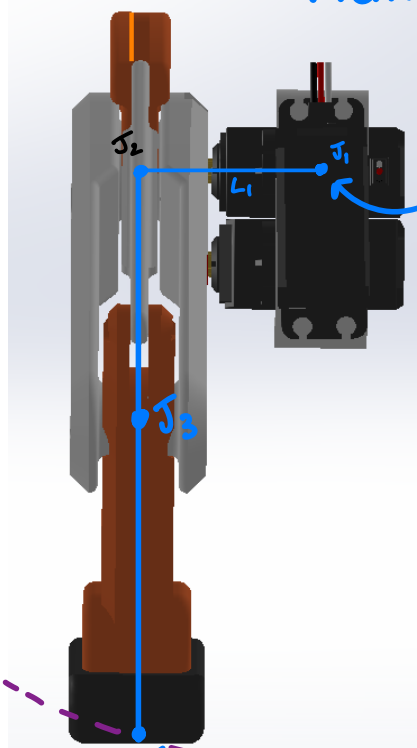
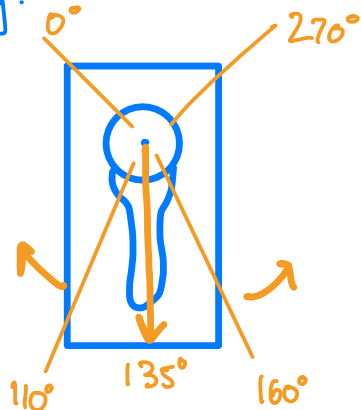
"desired foot contact point"

Do I need to get this complex??

Hip Joint :

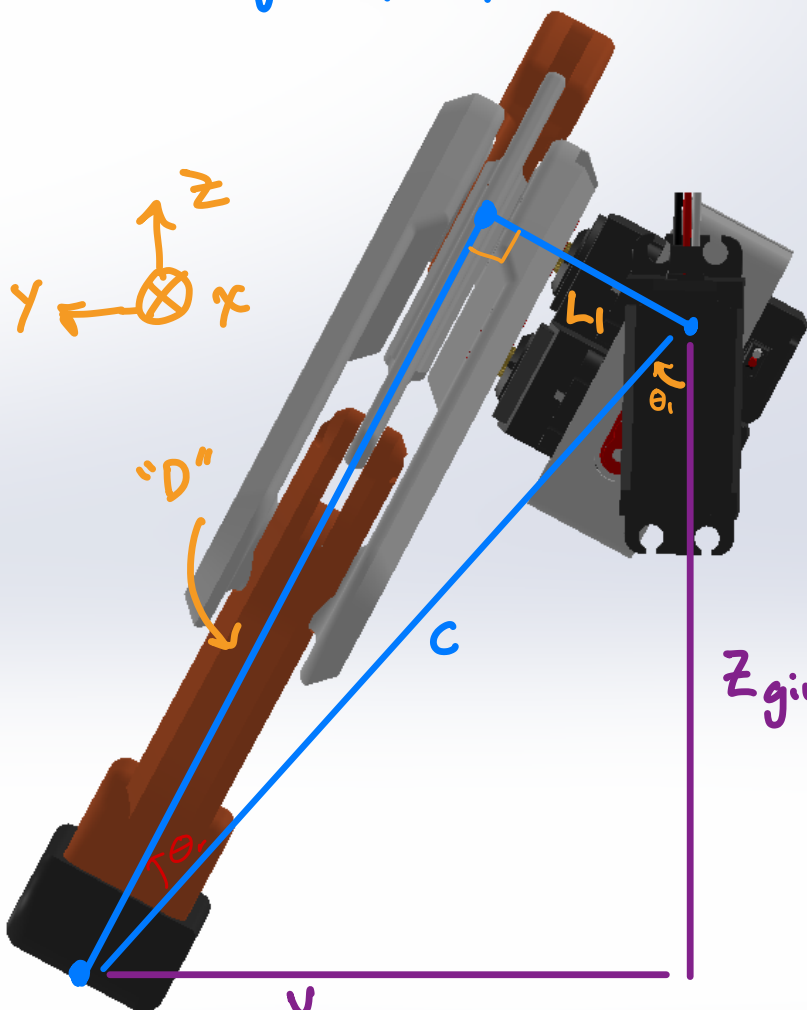
Back view of front left leg

Front view of hip servo on front left leg:



foot will trace path along this circle when hip rotates

When hip is angled up θ_1



D^* $\Rightarrow$ leg length desired in the leg plane

Goal: Find θ_1, D .

$$\frac{D}{\Rightarrow} C = \sqrt{Y_{given}^2 + Z_{given}^2}$$

$$D = \sqrt{C^2 - L_1^2}$$

$$\Rightarrow D = \sqrt{Y_{given}^2 + Z_{given}^2 - L_1^2}$$

$\rightarrow$ for vertical leg $\Rightarrow Y_{given} = L_1$

For θ_1

$$\theta_1 = \tan^{-1}(Y_{given} / Z_{given})$$

For right leg, Y_{given} flips $\Rightarrow \theta_1 = -\tan...$

This θ_1 is a virtual angle. For servo,

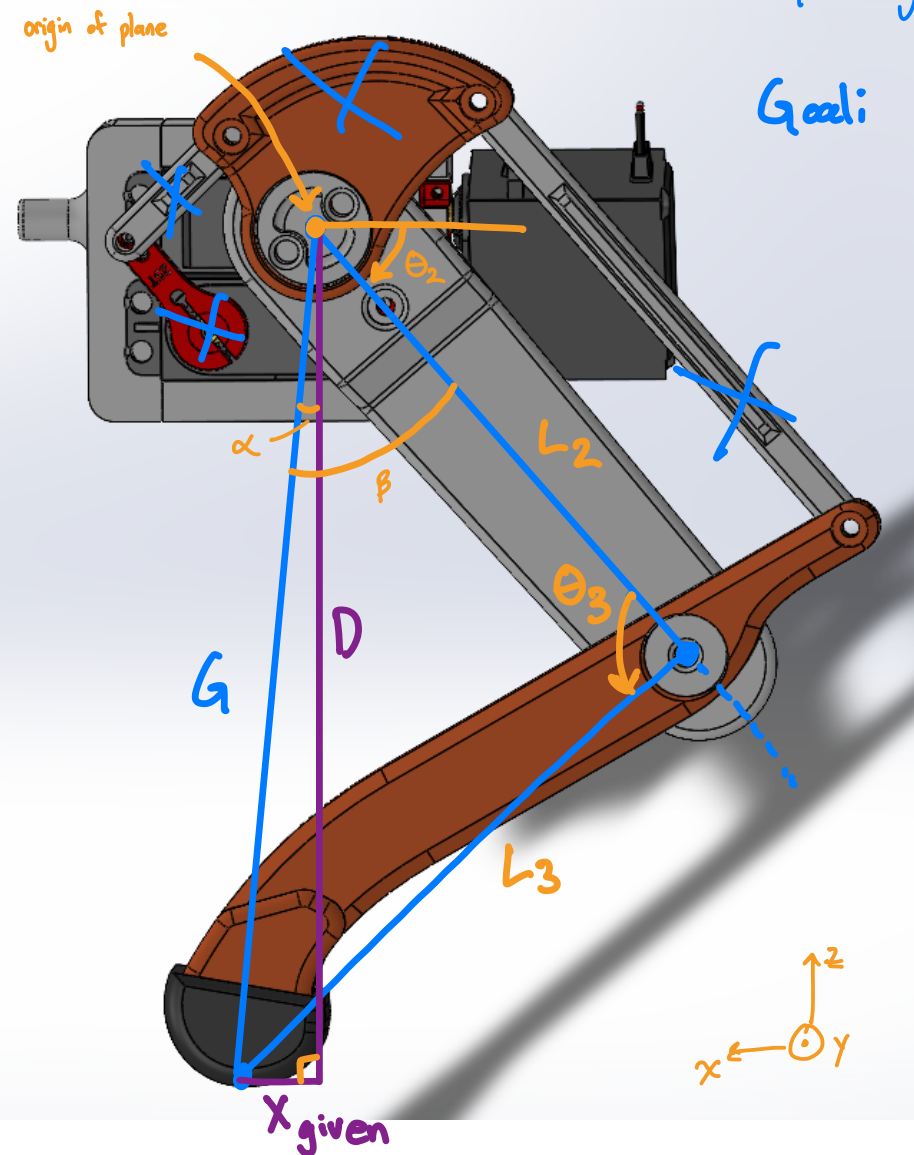
$$\theta_{hip servo} = 135^\circ + \theta_1$$

$$\theta_r = \tan^{-1}(L_1 / D)$$

$$\rightarrow \text{Actual } \theta_1' = \theta_1 - \theta_r$$

B/c when $\theta_1 = 0^\circ$, $\theta_r \neq 0$, need to remove this extra θ_r or it will be counted as an angle that needs to be set. $\uparrow$ to align servo 0° w/ math

Leg Plane (X-Z plane) : Simplify first, by not accounting for $\theta_{\text{offset}} = 5.88^\circ$ and pretending tibia servo is mounted on knee joint.



Goal: Find θ_2, θ_3 :

$$\rightarrow G = \sqrt{D^2 + X_{\text{given}}^2}$$

For θ_3 , cos law:

$$G^2 = L_2^2 + L_3^2 - 2L_2L_3 \cos \theta_3$$

$$\Rightarrow \theta_3 = \cos^{-1} \left(\frac{G^2 - L_2^2 - L_3^2}{-2L_2L_3} \right)$$

For θ_2 :

$$\alpha = \tan^{-1} \left(\frac{X_{\text{given}}}{D} \right)$$

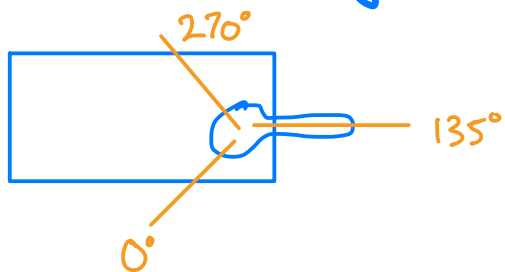
B. Sine law $\frac{\sin \beta}{L_3} = \frac{\sin \theta_3}{G}$

$$\Rightarrow \beta = \sin^{-1} \left(\frac{L_3 \sin \theta_3}{G} \right)$$

$$\Rightarrow \theta_2 = \frac{\pi}{2} - (\beta - \alpha)$$

θ_2, θ_3 are "virtual angles" $\rightarrow$ for servo,

Femur Servo Mounting / Range of Motion:

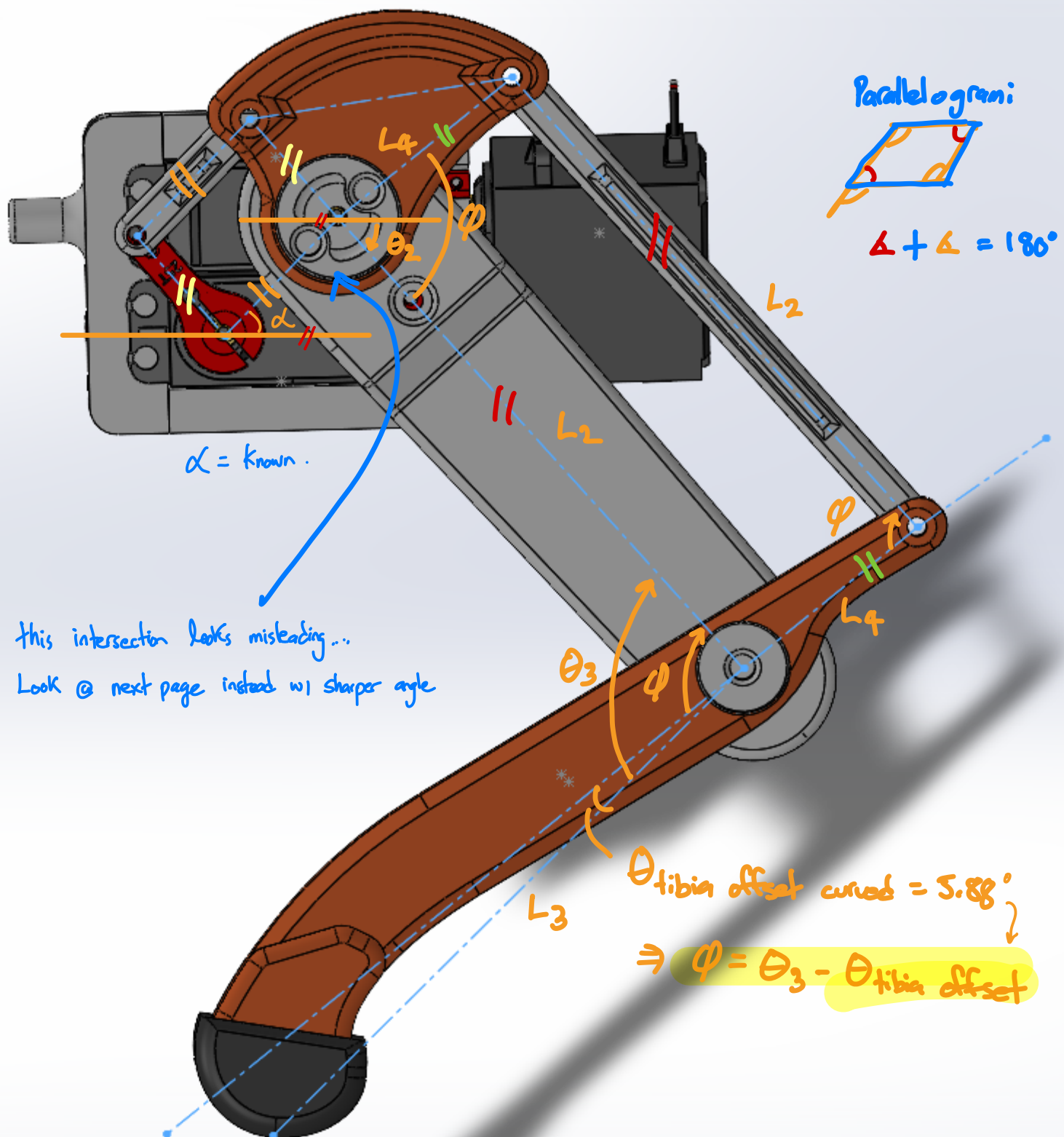


$$\Rightarrow \theta_{\text{femur servo}} = 135^\circ - \theta_2$$

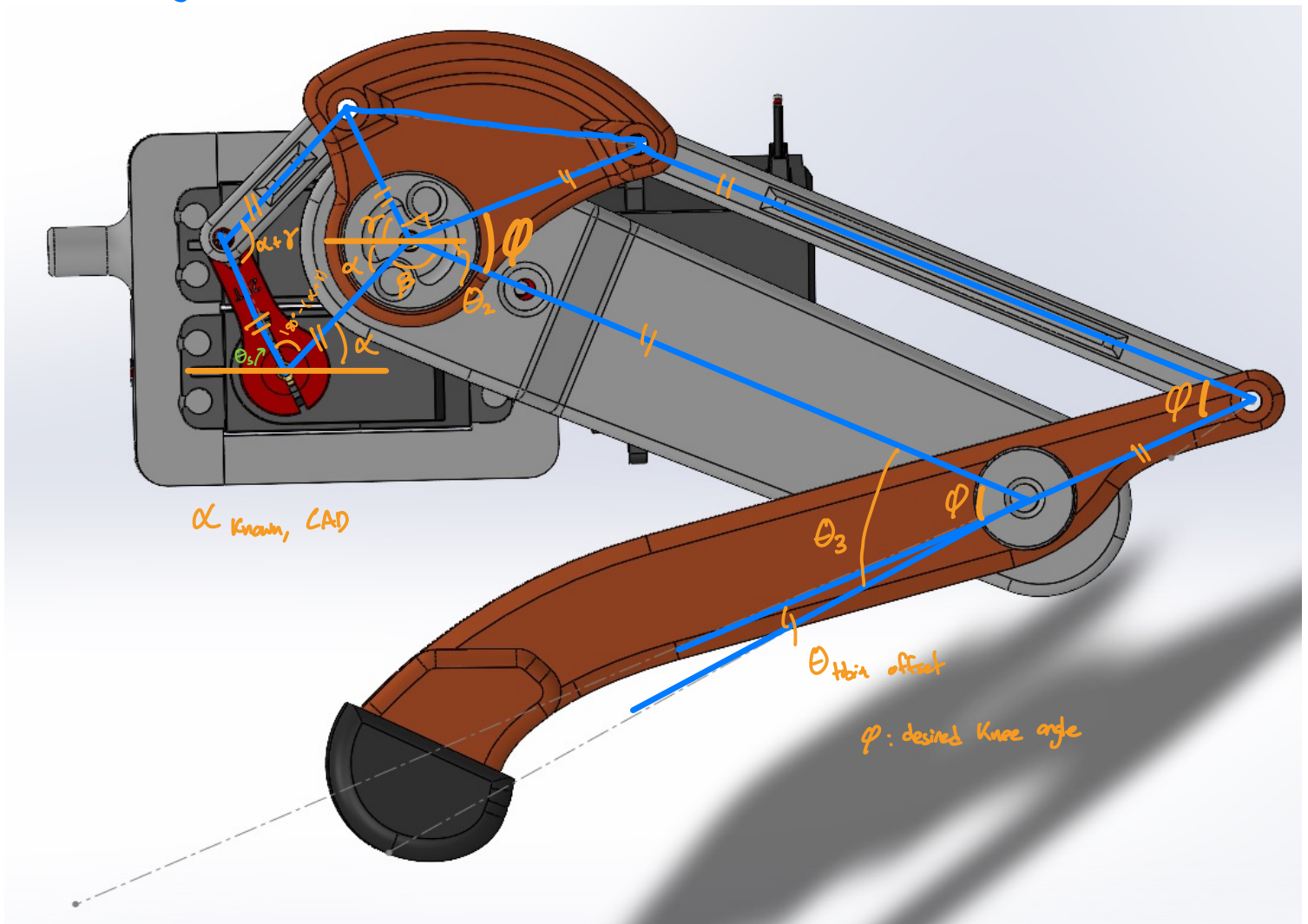
↑ offset alignment, varies wr each servo

For θ_{tibia} servo, need to draw linkages....

// = parallel pairs $\Rightarrow$ linkages are designed to be 2 consecutive parallelograms



Analyzing sharper angle for $\Theta_{\text{servo tibia}}$



$$\rightarrow \beta = 180 - \alpha - \theta_2$$

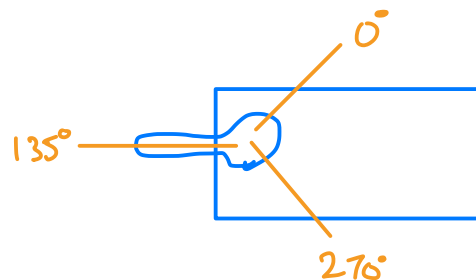
$$\rightarrow \gamma = 360 - 90 - \phi - \beta - \alpha$$

$$\begin{aligned} \therefore \theta_s &= 180^\circ - \alpha - [180^\circ - (\alpha + \gamma)] \\ &= -\alpha + \alpha + \gamma \\ &= \gamma \\ &= 360^\circ - 90 - \phi - \beta - \alpha \quad [\text{Subst. } \gamma] \\ &= 360^\circ - 90^\circ - \phi - \alpha - (180^\circ - \alpha - \theta_2) \quad [\text{Subst. } \beta] \\ &= 360^\circ - 90^\circ - \phi - \alpha - 180^\circ + \alpha + \theta_2 \\ &= 90^\circ - \phi + \theta_2 \end{aligned}$$

Now, to account for servo alignment:

$$\Theta_{\text{servo, tibia}} = 135^\circ - \theta_s$$

↑
offset
alignment



or possibly $= \phi + \theta_2$ based on other sources??